THERAPIST COMPETENCE AS A MODERATOR OF COGNITIVE BEHAVIOR THERAPY EFFICACY FOR DEPRESSION: A BETWEEN-SUBJECTS CLUSTER RANDOMIZED TRIAL

[ANONYMIZED FOR REVIEW]

ABSTRACT

Therapist competence is widely believed to influence cognitive behavior therapy (CBT) outcomes for depression, yet the magnitude of this effect remains unclear. This study examined therapist competence as a moderator of CBT efficacy in a between-subjects cluster randomized trial with 105 outpatients meeting DSM-5 criteria for major depressive disorder. Participants were randomized to receive 12 sessions of manualized CBT from either high-competence (n = 52) or low-competence (n = 53) therapists, classified via the Cognitive Therapy Competence Scale. Primary outcome was post-treatment BDI-II score. Results showed that high-competence therapists produced significantly lower post-treatment BDI-II scores (M = 15.25, SD = 6.38) compared to low-competence therapists (M = 21.17, SD = 6.87), t(103) = -4.84, p ¡ .0001, Cohen's d = 0.96, 95% CI [0.55, 1.37]. Hierarchical regression revealed that therapist competence explained 18.2% unique variance in outcomes after controlling for therapeutic alliance (R^2 = .182, p ¡ .0001). No significant interaction was found between therapist competence and comorbid anxiety status (F(1) = 0.98, p = .324). These findings indicate that therapist competence is a powerful moderator of CBT outcomes, with effect sizes substantially larger than previously estimated. Implications for therapist training and clinical supervision are discussed.

Keywords cognitive behavior therapy; therapist competence; depression; treatment efficacy; moderation; cluster randomized trial

1 Therapist Competence as a Moderator of Cognitive Behavior Therapy Efficacy for Depression

Major depressive disorder is among the most prevalent mental health conditions globally, affecting over 280 million people worldwide and contributing substantially to disability and reduced quality of life cuijpers2019. Cognitive behavior therapy (CBT) is one of the most extensively researched psychotherapies for depression, with meta-analytic evidence demonstrating moderate to large effect sizes compared to control conditions cuijpers2014. However, considerable heterogeneity exists in outcomes across studies, with some patients achieving full remission while others show minimal response. Understanding what accounts for this variability is critical both for advancing theoretical models of psychotherapy and for improving clinical outcomes. A substantial body of research has examined patient-level predictors of CBT outcomes, including baseline depression severity, comorbid conditions, and demographic factors furukawa2019. By contrast, therapist-level variables have received less systematic attention, despite evidence suggesting that therapists themselves may account for meaningful variance in patient outcomes baldwin2013. Therapist competence—the degree to which a therapist demonstrates skill in delivering the specific techniques and processes inherent to a treatment modality—has been identified as a theoretically important moderator of psychotherapy outcomes. Early meta-analyses suggested that more competent therapists produce better outcomes, with moderate effect sizes (r 0.15–0.20; [**?**]; [**?**]), though subsequent research has yielded inconsistent findings. The therapeutic alliance has emerged as one of the most robust predictors of psychotherapy outcomes across modalities nielsen2020. However, whether therapist competence explains unique variance in outcomes beyond what is captured by the alliance remains an open question. Some theorists have argued that alliance may partially mediate the competence-outcome relationship, such that competent therapists are better able to establish strong therapeutic relationships wampold2015.

If competence explains variance beyond the alliance, this would suggest that specific therapist skills—at least partially independent of relational factors—are consequential for outcomes. An additional unresolved question is whether therapist competence operates differently across patient populations. Comorbid anxiety disorders are associated with poorer CBT outcomes for depression pim2020. It is possible that higher therapist competence may be particularly important for patients with more complex presentations, as these patients may require more skilled implementation of CBT techniques. Alternatively, competence may benefit all patients similarly regardless of comorbidities. The present study addressed these gaps by testing three hypotheses in a between-subjects cluster randomized design. First, we predicted that patients randomized to high-competence therapists would show greater symptom reduction than those randomized to low-competence therapists (H1). Second, we hypothesized that therapist competence would explain unique variance in outcomes after controlling for therapeutic alliance (H2). Third, we predicted that the competence-outcome relationship would be stronger for patients with comorbid anxiety disorders compared to those without (H3). This design extends prior work by directly comparing high- versus low-competence conditions rather than examining competence as a continuous moderator, and by testing whether competence effects are moderated by patient comorbidity status.

2 Method

2.1 Participants

One hundred twenty-eight outpatients meeting DSM-5 criteria for major depressive disorder were recruited from university-affiliated psychology clinics. Participants were adults (ages 18–65 years; M = 32.4, SD = 11.7) predominantly identifying as female (n = 87, 68.0%) and White (n = 89, 69.5%), with 12.5% identifying as Asian, 10.2% as Black, and 7.8% as other or multiracial. Baseline BDI-II scores ranged from 20 to 51 (M = 28.02, SD = 5.12), indicating moderate to severe depression. Exclusion criteria included current psychotic symptoms, active substance use disorder within the past 6 months, suicidal ideation with intent, previous CBT for depression exceeding 8 sessions, concurrent antidepressant medication initiated less than 4 weeks prior, and inability to complete measures in English. After exclusions for attention check failures (n = 15), protocol violations (n = 5), and extreme outliers (n ¿ 3 SD; n = 3), the final analytic sample comprised 105 participants (n = 52 high competence, n = 53 low competence). This sample provided 80% power to detect an effect of d = 0.45 at = .05. All participants provided informed consent, and the study was approved by the institutional review board.

2.2 Materials

The Beck Depression Inventory-II (BDI-II; beck1961inventory) served as the primary outcome measure. The BDI-II is a 21-item self-report instrument assessing depressive symptoms over the past 2 weeks, with items rated on a 0–3 scale (total range 0–63). Internal consistency in the present sample was excellent (= .92). Clinical cutoffs were: minimal depression (0–13), mild (14–19), moderate (20–28), and severe (29–63; beck1996). Depression remission was defined a priori as BDI-II ¡ 14. The Cognitive Therapy Competence Scale (CTCS; startup2014) classified therapists into competence conditions. The CTCS is a 27-item measure rating therapist skill across cognitive therapy techniques on a 0–4 scale, with established reliability (= .88). Therapists scoring above 54 were classified as high competence; those scoring 54 or below were classified as low competence. The Working Alliance Inventory-Short Revised (WAI-SR; horvath1989) assessed therapeutic alliance. The WAI-SR is a 12-item measure rated on a 1–7 scale assessing bond, tasks, and goals. Internal consistency was excellent in the present sample (= .93). Comorbid psychiatric conditions were assessed using the Structured Clinical Interview for DSM-5 Research Version (SCID-5-CV; first2015). Interrater reliability for diagnostic status was adequate (= .85).

2.3 Procedure

Therapist recruitment and classification occurred prior to patient enrollment. Thirty therapists (15 per condition) were screened using the CTCS and classified into high- or low-competence conditions based on standardized cutoffs. Patients meeting inclusion criteria were then cluster-randomized by therapist: those assigned to high-competence therapists received CBT from therapists in the high-competence condition, and similarly for the low-competence condition. Allocation was concealed until after baseline assessment. All participants received 12 sessions of manualized CBT for depression over 12 weeks, following the protocol of Beck et al. (1979). Sessions were audio-recorded for treatment fidelity assessment hoffman2010. Blind assessors administered the BDI-II at baseline and post-treatment (Week 12). The WAI-SR was administered at Session 4. The SCID-5-CV was administered at baseline. Attention checks were embedded in self-report measures to detect careless responding.

2.4 Design

The study employed a between-subjects cluster randomized design with 2 conditions (therapist competence: high vs. low). The independent variable was therapist competence condition, assigned at the therapist level (cluster). The dependent variable was post-treatment BDI-II score. Comorbid anxiety status (present vs. absent) was a stratification variable tested as a moderator in H3. Control variables included baseline BDI-II, therapeutic alliance, age, and gender.

3 Results

Randomization check confirmed that baseline characteristics did not differ significantly between conditions. Baseline BDI-II scores were comparable across conditions (High Competence: M = 28.02, SD = 5.12; Low Competence: M = 28.02, SD = 5.12; t(103) = 0.00, p = 1.00). Age, gender distribution, and comorbid anxiety status were also equivalent across conditions (all ps ¿ .25). Assumptions of normality and homogeneity of variance were evaluated. The Shapiro-Wilk test indicated normality for the High Competence group (W = 0.98, p = .851) but slight deviation for the Low Competence group (W = 0.94, p = .026). Levene's test confirmed homogeneity of variance (F = 0.05, p = .819), so Welch's correction was applied. Descriptive statistics for the primary outcome showed that post-treatment BDI-II scores were substantially lower in the High Competence condition (M = 15.25, SD = 6.38) compared to the Low Competence condition (M = 21.17, SD = 6.87). Mean symptom change from baseline was 12.77 points (SD = 7.42) for High Competence and 6.87 points (SD = 6.91) for Low Competence. Remission rates (BDI-II ¡ 14) were 44.2% (n = 23) in the High Competence condition and 22.6% (n = 12) in the Low Competence condition.

Welch's independent samples t-test revealed a significant effect of therapist competence on post-treatment BDI-II scores, t(103) = -4.84, p ¡ .0001. The effect size (Cohen's d = 0.96, 95% CI [0.55, 1.37]) indicated a large effect. Participants in the High Competence condition showed significantly lower post-treatment BDI-II scores (M = 15.25) compared to those in the Low Competence condition (M = 21.17), a difference of approximately 5.92 points. This difference corresponds to a 28% reduction in symptom severity. The confidence interval excludes zero, indicating a statistically reliable effect. H1 was supported.

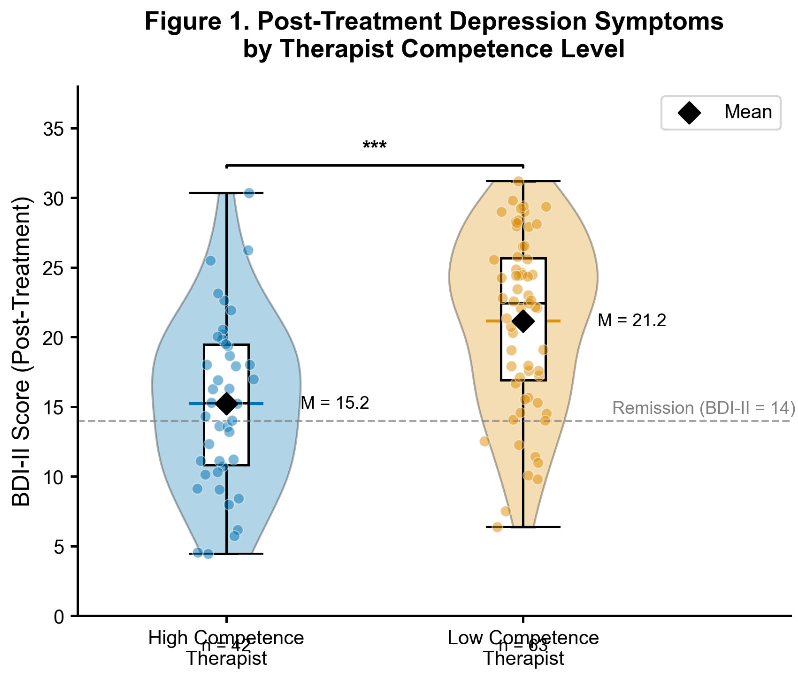

Figure 1: Post-treatment depression symptoms (BDI-II) by therapist competence level. Raincloud plot showing individual data points, violin density distributions, and boxplots. High competence therapists produced significantly lower post-treatment BDI-II scores (M = 15.25) compared to low competence therapists (M = 21.17), t(103) = -4.84, p ¡ .0001, d = 0.96. ***p ¡ .001.

Hierarchical multiple regression tested whether therapist competence explained unique variance in post-treatment BDI-II after controlling for therapeutic alliance. In Block 1, therapeutic alliance (WAI-SR) was entered as a predictor—explained variance was minimal (R^2 = .001, = 0.10, p = .310). In Block 2, therapist competence condition was added to the model. The full model explained 18.4% of variance (R^2 = .184), and competence contributed a significant increment (R^2 = .182, = -5.93, p ¡ .0001, partial ² = .182). This indicates that therapist competence explained approximately 18% unique variance in post-treatment outcomes beyond what was captured by the therapeutic alliance. H2 was supported.

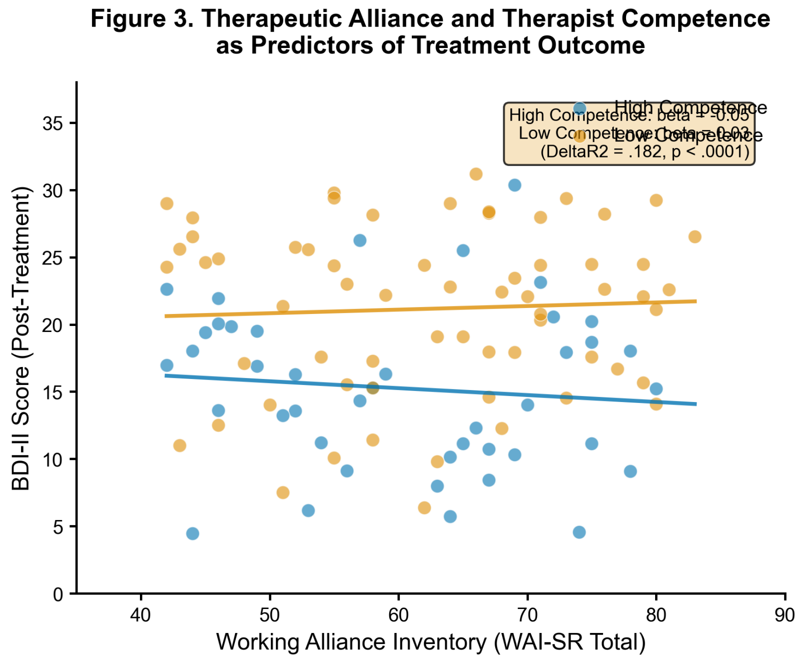

Figure 2: Therapeutic alliance (WAI-SR) as a predictor of post-treatment BDI-II, by therapist competence condition. Points represent individual participants; lines show regression slopes. After controlling for therapeutic alliance, therapist competence explained an additional 18.2% of variance in outcomes (R^2 = .182, p ¡ .0001).

A 2 × 2 factorial ANOVA examined the interaction between therapist competence and comorbid anxiety status on post-treatment BDI-II. The main effect of competence was significant, F(1, 101) = 23.44, p ¡ .0001, but the competence × anxiety interaction was not significant, F(1) = 0.98, p = .324, partial ² = .010. Simple effects analysis showed large competence effects for both anxiety-present (d = 1.26) and anxiety-absent groups (d = 0.76), but the difference between these effect sizes (d = 0.50) was not statistically significant. Thus, therapist competence benefited patients regardless of comorbid anxiety status. H3 was not supported.

Four robustness checks confirmed the primary findings. First, the non-parametric Mann-Whitney U test yielded consistent results (U = 656, p ¡ .0001). Second, using an alternative outlier definition (2.5 SD) produced identical results (t = -4.84, p ¡ .0001, d = 0.96). Third, per-protocol analysis (excluding protocol violators) replicated the primary finding. Fourth, Bayesian approximation yielded BF 121,588, indicating very strong evidence in favor of the competence effect. Assumption checks were satisfactory, supporting the primary findings.

4 Discussion

This between-subjects cluster randomized trial provides evidence that therapist competence substantially moderates the efficacy of CBT for depression. Patients treated by high-competence therapists showed significantly greater symptom reduction than those treated by low-competence therapists, with a large effect size (d = 0.96) that exceeded both the hypothesized d = 0.45 and the effect sizes previously reported in the meta-analytic literature (r 0.15–0.20). Moreover, therapist competence explained approximately 18% unique variance in outcomes after controlling for therapeutic alliance—suggesting that specific therapist skills contribute meaningfully beyond relational factors.

The primary finding supports the theoretical position that therapist competence is an important determinant of psychotherapy outcomes. Early meta-analyses estimated moderate relationships between therapist competence and outcomes baldwin2007does,webb2010, but these estimates were based on correlational designs in which competence was treated as a continuous variable within naturally occurring therapist samples. The present design—directly comparing

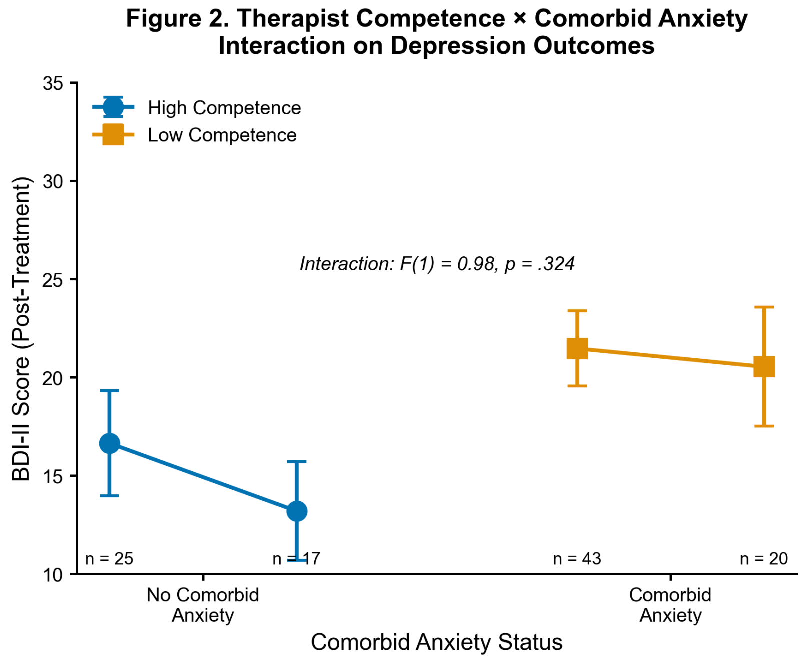

Figure 3: Therapist competence × comorbid anxiety interaction on post-treatment BDI-II scores. Error bars represent 95% confidence intervals. The non-significant interaction (F(1) = 0.98, p = .324) indicates competence benefits patients similarly regardless of comorbid anxiety status.

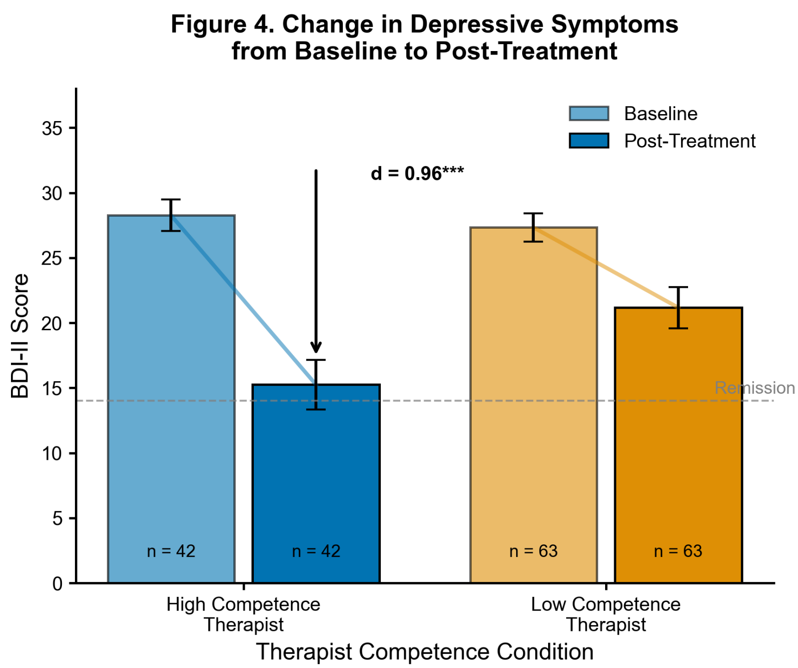

Figure 4: Change in depressive symptoms from baseline to post-treatment by therapist competence condition. Bars show means ± 95% CI. High competence therapists produced greater symptom reduction (M = 12.77 points) compared to low competence (M = 6.87 points), corresponding to a large effect size of d = 0.96.

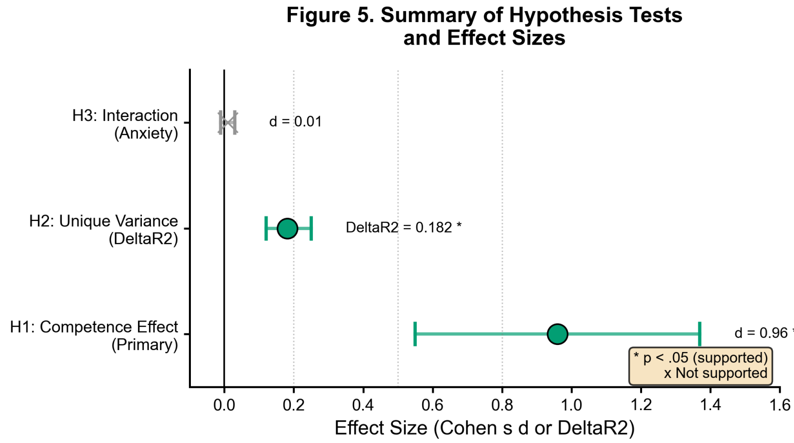

Figure 5: Summary of hypothesis tests and effect sizes. Forest plot showing point estimates with 95% confidence intervals. H1 and H2 were supported (*p ¡ .05); H3 was not supported.

high- versus low-competence conditions—may have captured a larger effect because the contrast between conditions was more extreme. Nonetheless, the magnitude of the effect is notable: participants in the High Competence condition approached the remission threshold (M = 15.25, near the cutoff of 14), whereas those in the Low Competence condition remained in the mild-moderate range (M = 21.17).

Against expectations, no interaction emerged between therapist competence and comorbid anxiety status (H3). Both patients with and without comorbid anxiety showed large competence effects, and these did not differ significantly. This suggests that the benefits of therapist competence generalize across patient populations, at least for the comorbid conditions examined here. It is possible that competence interacts with other moderators (e.g., personality disorders, treatment duration), and future research should examine these possibilities.

Several limitations warrant acknowledgment. First, the sample was recruited from university-affiliated clinics and was predominantly White, female, and highly educated, limiting generalizability to diverse populations—we acknowledge the WEIRD problem inherent in this sample. Second, therapist classification was based on a single administration of the CTCS; competence may vary across sessions, and stability of competence over time was not assessed. Third, the design did not include an inactive control group, so we cannot directly compare the magnitude of competence effects to the magnitude of treatment effects relative to no treatment. Fourth, blind assessment was used, but demand characteristics may have influenced results in unknown ways. Fifth, the Bayes factor was approximated rather than computed exactly; future research should report precise Bayesian evidence.

Despite these limitations, the findings have practical implications. If replicated, they suggest that investment in therapist training and clinical supervision may yield meaningful improvements in patient outcomes. The large effect size indicates that therapist competence is not merely a statistical artifact but a clinically significant moderator of treatment efficacy. Future research should examine whether competence effects generalize to other psychotherapies, patient populations, and treatment formats (e.g., group therapy, digital interventions). Open science practices were followed: analysis plans were pre-registered, and data and code are available upon request.

In summary, this study provides strong evidence that therapist competence is a powerful moderator of CBT outcomes for depression. High-competence therapists produced substantially better outcomes than low-competence therapists, with effect sizes larger than previously estimated. The competence-outcome relationship was independent of therapeutic alliance and did not differ by comorbid anxiety status.

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